

Graphene and 2D Materials: Science, Processing, and Applications (ChE 494)

Quiz 3: Science, Processing and Characterization of 2D Materials

November 26, 2019

60 Minutes

Total: 100

1. Which one of the following is NOT a sp^2 hybridized material?

- ☒ (A) Graphite (B) Hexagonal Boron Nitride (C) Graphene (D) Diamond

2. Layered materials consist of 2D platelets weakly stacked to form 3D structures. Therefore, it is possible to physically exfoliate 2D crystals from their 3D counterpart. Which of the following 3D bulk material cannot be physically exfoliated into its 2D structure?

- (A) Graphite (B) Bulk Hexagonal Boron Nitride (C) Bulk MoS_2 ☒ (D) Silicon

3. Graphene growth mechanism *via* CVD: what is the key difference between copper and nickel?Answer: Carbon Affinity: $Cu < Ni$

Growth mechanism: Cu - surface aggregation

Ni - Segregation & precipitation.

4. Graphene is the 2D building block for sp^3 carbon allotropes of every other dimensionality.

- (A) True ☒ (B) False

5. In case of Stone-Wales defect, the defected structure retains the same number of atoms as pristine graphene, and no dangling bonds are introduced.

- ☒ (A) True (B) False

6. Which one from the following materials will have less sp^2/sp^3 ratio of hybridized carbon? C

- ☒ (A) Reduced Graphene Oxide (B) Graphene ☒ (C) Graphene Oxide

7. Which of the following polymers can be used for the transfer of graphene?

- (A) Polymethyl methacrylate (B) Polydimethylsiloxane ☒ (C) Both

8. Transition metals have partially-filled d-orbitals that enable them to adsorb and interact with the hydrocarbons that are used as the carbon sources. Arrange the four transition metals (Fe, Ni, Co, and Cu) in 'decreasing' carbon solubility.

Answer: Fe, Co, Ni, Cu

9. Which of the following chemicals can be used for etching of the Cu foil for the transfer of graphene?

- (A) Ammonium persulfate (B) Ferric chloride (C) Copper chloride ☒ (D) All

10. The graphene synthesis via chemical exfoliation route follows: graphite powder oxidation..... graphite oxide sonication..... graphene oxide reduction..... ~~fax~~..... reduced graphene oxide.

11. The graphene growth rate depends on the specific polar SiC crystal face: graphene forms much slower on the silicon-terminated face (0001) surface (or Si-face) than on the carbon-terminated face (000-1) surface (or C-face).

☒ (A) True

☒ (B) False

12. In case of the confinement-controlled sublimation (CCS) method: sublimed Si gas is confined in a graphite enclosure so that growth occurs in near thermodynamic equilibrium. Growth rate is controlled by the enclosure aperture (leak), and the background gas pressure.

☒ (A) True

(B) False

13. From the following, which graphene will have highest charge carrier mobility ($\text{cm}^2\text{V}^{-1}\text{s}^{-1}$) and why?

☒ (A) CVD

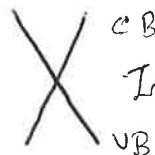
(B) Mechanically Exfoliated

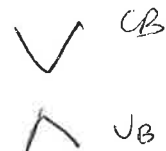
(C) Chemically Exfoliated

Explain: CVD will have highest charge carrier mobility because mainly it is a bottom up approach. It starts from single atoms. ~~the~~ other processes might affect the mobility because it might inherit some properties from its bulk structure.

14. Draw the band structure (E-K diagram) of single-layer graphene with intrinsic, n-type doping, and p-type doping characteristics.

Answer:

 CB
Intrinsic
VB

 CB
p-type
VB

15. Out of the following: what is the suitable substrate for graphene devices or electronics and why?

(A) Silicon Dioxide-Coated Silicon (SiO_2/Si)

☒ (B) Hexagonal Boron Nitride (h-BN)

Explain: SiO_2/Si will be a complete insulator where as h-BN as a substrate will make a semi-conductor device on applied potential whereas SiO_2/Si substrate will remain unaffected.

16. For the synthesis of 'white graphene' via CVD, the crystalline solid precursor (ammonia borane) is thermally decomposed to at temperature of $60\sim 180^\circ\text{C}$.

Answer: NH_3 , N_2 , H_2 , B

17. MoS_2 exhibits layer-dependent properties: Transition from a direct band gap in the bulk ($E_g=1.2\text{ eV}$) to an indirect band gap in the monolayer ($E_g=1.8\text{ eV}$).

(A) True

☒ (B) False

18. In case of graphene synthesis on metal catalyst substrates *via* CVD, graphene crystallites will seed at positions of locally 'lower chemical potential'. What are those positions in the metal catalyst (for example: copper)?

Answer: Grain boundaries, Contaminants

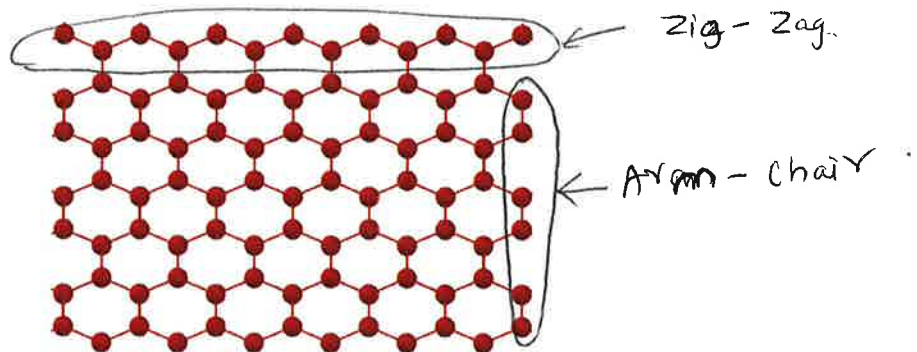
19. Hexagonal boron nitride (h-BN) is isostructural and isoelectronic to graphene. How much lattice mismatching is between graphene and h-BN?

Answer: 1.7%

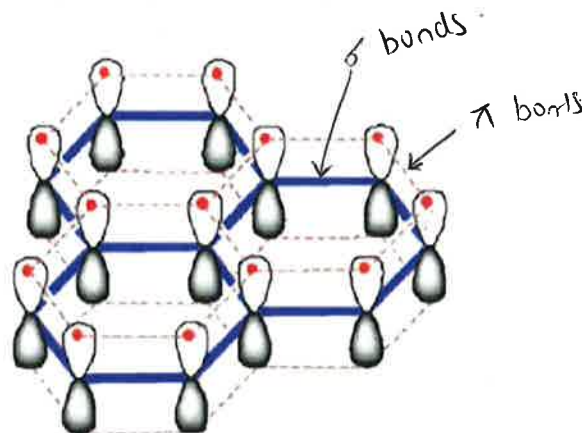
20. What does 2H, 3R, and 1T polytypes of MoS_2 represent (which symmetry, how many layers per repeat unit, and which coordination)?

Answer: 2H: Hexagonal - 2 layers
3R: Rhombohedral - 3 layers
1T: Trigonal bipyramidal - 1 layer

21. Graphene edges control the optical, magnetic, and electronic properties of graphene. In particular, termination, chemical functionalization, and reconstruction of graphene edges leads to crucial changes in the properties of graphene, so control of the edges is important to the development of applications in electronics and optoelectronics. There are two types of graphene edges: 'zig-zag' and 'arm-chair' edges. Where are those edges in the following single-layer graphene honeycomb lattice?



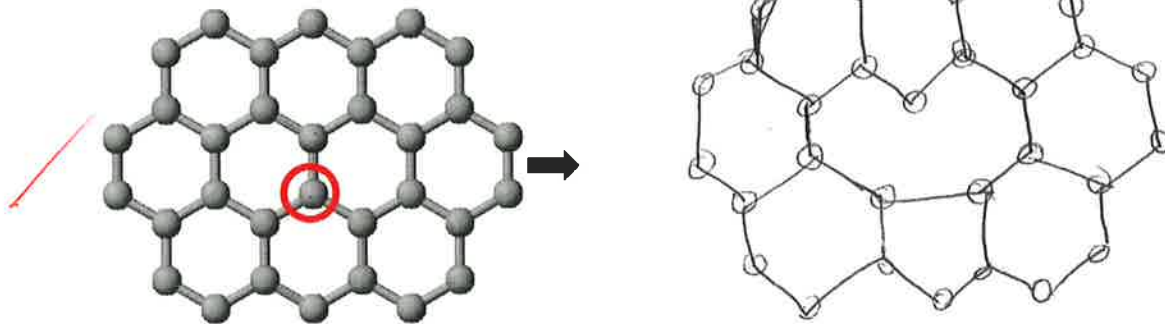
22. In graphene, the sigma (σ) bonds are localized electrons and pi (π) bonds are delocalized electrons. In the following figure, designate the localised ' σ ' and delocalised ' π ' bonds in the graphene lattice.



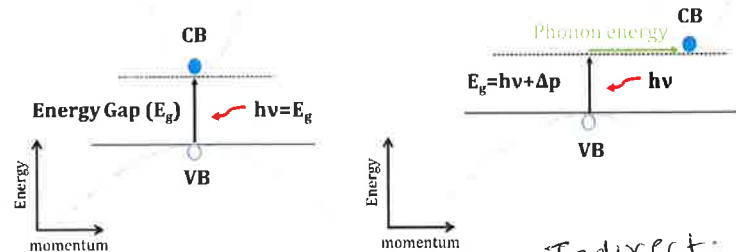
Name:

UIN:

23. Single vacancies undergoes a Jahn-Teller distortion which leads to the saturation of two of the three dangling bonds toward the missing atom. Draw the final structure after you make a single vacancy by removing the carbon atom (circled red) in the following honeycomb graphene lattice.

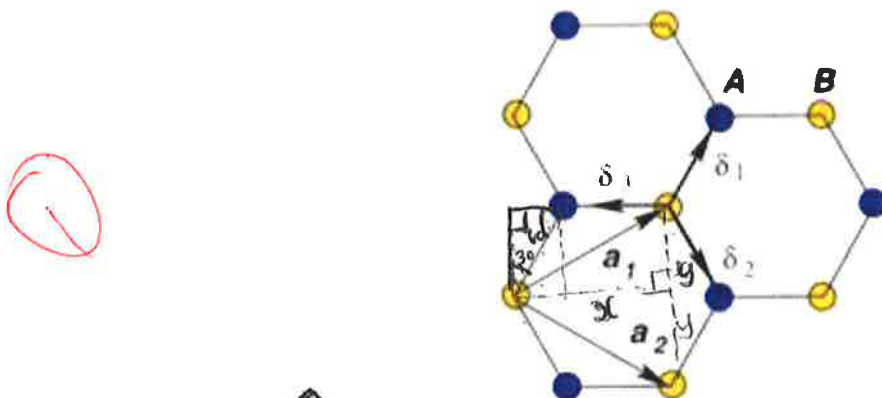


24. From the following figure: Differentiate between direct and indirect bandgap semiconductors and explain the reason.

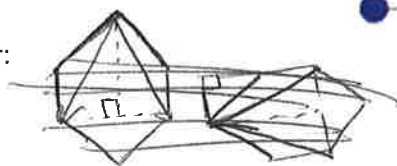


Explain:

- Direct**
The e^- can jump from VB to CB in one go
No change in crystal momentum
- Indirect**
The e^- has to jump to a virtual state first & then to the CB.
Crystal momentum changes.
25. Determine the lattice unit vectors (a_1 and a_2) in the following honeycomb graphene lattice.



Answer:



$$a_1 = x + y$$

$$a_2 = x - y$$

$$y = a \cos 60^\circ$$

$$x = a \sin 60^\circ$$

$$x = a \sin 30^\circ + a$$

$$= a(\sin 30^\circ + 1)$$

$$\therefore a_1 = a(\sin 30^\circ + 1) + a \cos 60^\circ$$

4 | Page

$$a_2 = a(\sin 30^\circ + 1) - a \cos 60^\circ$$

Name:

UIN:

26. X-ray photoelectron spectroscopy (XPS) is a surface-analysis spectroscopic technique, which measures the.....

- (B) Topography (B) Morphology ☒ (C) Elemental Composition (D) All

27. Raman spectroscopy bands arise from a change in the dipole moment (μ) of the molecule due to an interaction of light with the molecule.

- (B) True ☒ (B) False

28. When the laser excitation frequency is close with an electronic absorption band of the sample: the intensity enhancement of the Raman scattered light can occur. What is this scattering called?

- (A) Surface-Enhanced Raman Scattering ☒ (B) Resonance Raman Scattering (C) All

29. Raman spectroscopy is a spectroscopic technique used to study vibrational, rotational, and other low-frequency modes in a system. It relies on scattering (Raman Scattering) of monochromatic light, usually from a laser in the visible, near infrared or near ultraviolet range.

- (C) Elastic ☒ (B) Inelastic (C) Both Elastic and Inelastic

30. What kind of information we can get from the Raman spectroscopy of sp^2 nanocarbons?

- (D) Crystallite size (B) Doping (C) Defects ☒ (D) All

31. Scanning electron microscopy (SEM) can be employed for the following sample information.

- (B) Topography (B) Morphology (C) Compositional Information ☒ (D) All

32. What is the typical resolution of electron microscopes (SEM and TEM)?

- (B) 200 nm (B) 200 μm ☒ (C) 0.2 nm

33. Raman techniques are useful for graphene because the absence of a makes all wavelengths of incident radiation resonant, thus the Raman spectrum contains information about both atomic structure and electronic properties.

- ☒ (A) sp^3 hybridizations (B) Bandgap (C) Defects (D) All

34. Scanning tunneling microscopy (STM) is applicable when both the sample and tip are conducting.

- ☒ (B) True (B) False

35. The resultant dipole moment of a molecule is expressed in the following equation. State Rayleigh scattering, Raman Stokes scattering and Raman Anti-Stokes scattering frequencies/components.

$$\mu = [\alpha_0 E_0 \cos(2\pi \nu t)] + \frac{1}{2} (\delta\alpha/\delta Q)_0 E_0 [\cos(2\pi(\nu - \nu_{\text{vib}})t) + \overbrace{\cos(2\pi(\nu + \nu_{\text{vib}})t)}^{\text{Anti-Stokes}}]$$

Rayleigh Stokes

36. The Raman E_{2g} in-plane vibrational mode for h-BN is generally centers at.....

(A) 1560 cm^{-1}

☒ (B) $1365\text{--}1370\text{ cm}^{-1}$

(C) 2700 cm^{-1}

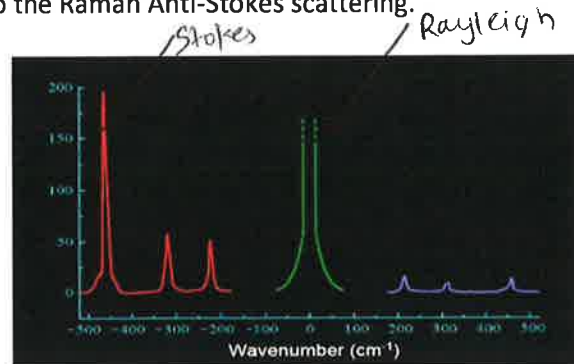
(D) 1350 cm^{-1}

37. Graphene's G-band is the first-order Raman spectrum, corresponds to the optical mode vibration of two neighboring carbon atoms on a sp^2 -hybridized graphene layer.

☒ (A) True

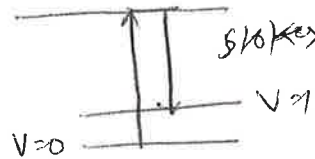
(B) False

38. State Rayleigh scattering, Raman Stokes scattering and Raman Anti-Stokes scattering from the following Raman spectrum. Explain the reason behind the higher intensity for Raman Stokes scattering in contrast to the Raman Anti-Stokes scattering.

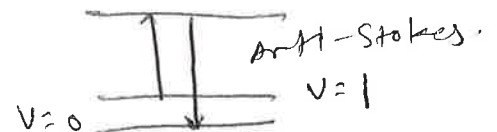


Explain:

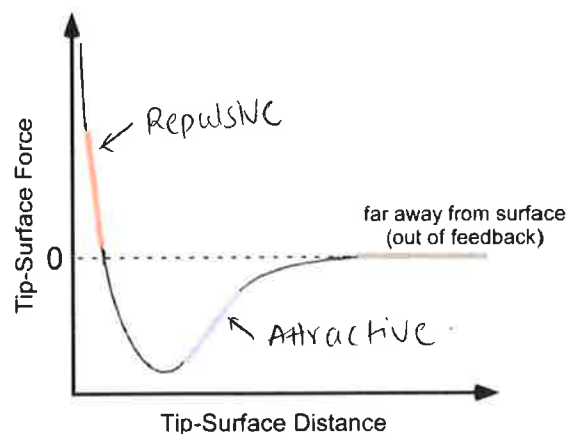
Raman Stokes has higher intensity because the scattered light is of higher intensity.



Raman anti-Stokes has lower intensity because the scattered light has lower intensity.



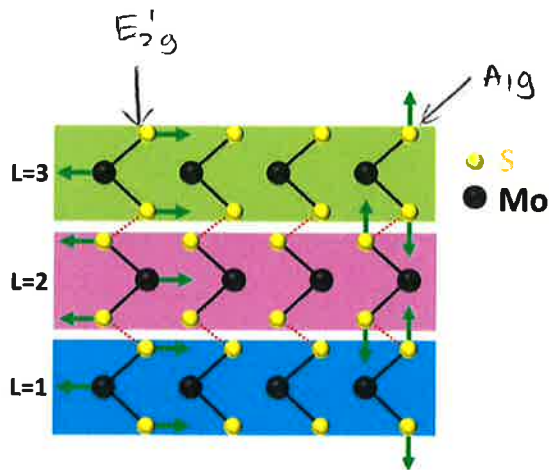
39. The typical van der Waals force curve for atomic force microscope (AFM) operation mode is presented in the figure below. Show various force-regime and mode of operations?



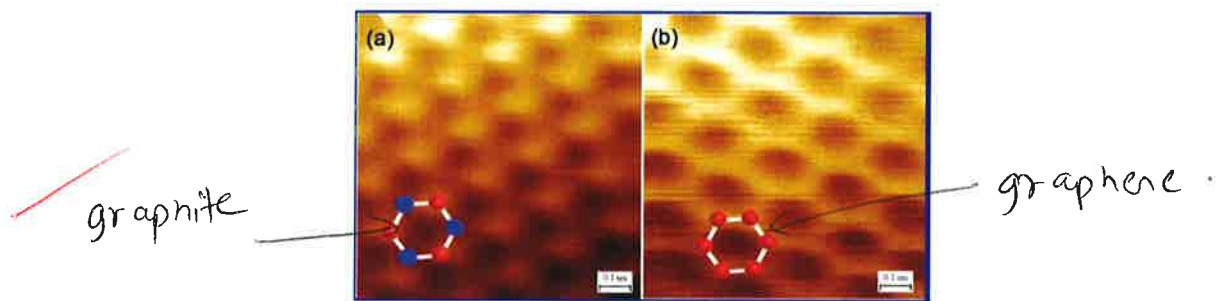
Name:

UIN:

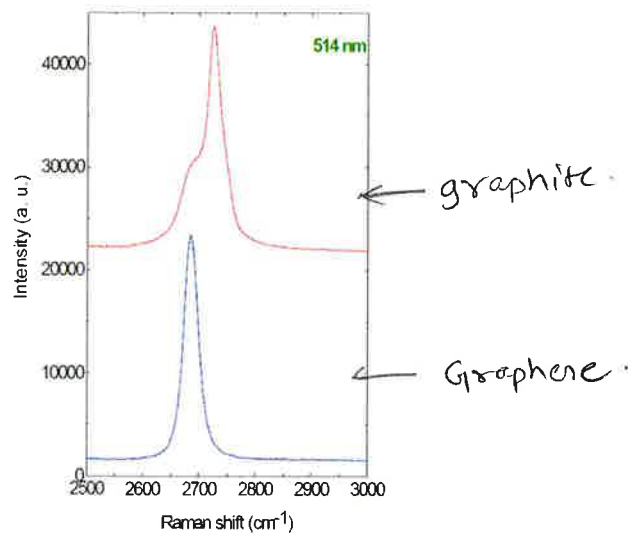
40. MoS_2 exhibits two Raman vibrational modes: E'_{2g} and A_{1g} . Designate these modes in the following figure.



41. Following figure shows the scanning tunnelling microscope (STM) images of carbon materials: 2D graphene and 3D graphite. Designate graphene and graphite in the following figure.



42. The Raman spectra for 2D-band of graphene and graphite is presented in the following figure. Differentiate between graphene and graphite.

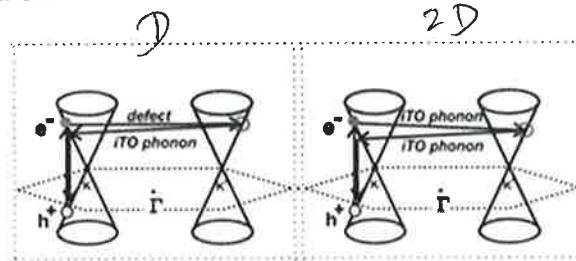


43. The frequencies, widths and intensities of the Raman E'_{2g} and A_{1g} peaks are strongly influenced by the thicknesses of the ultrathin MoS_2 flakes. Further, the frequencies of E'_{2g} and A_{1g} peaks can be used as key features to identify the layer number of a MoS_2 flake.

(A) True

(B) False

44. The double-resonance Raman process for D and 2D bands are presented in the figure below. Differentiate between them.



45. Scanning electron microscope (SEM) uses a focused probe to extract the structural and chemical information of the sample including 2D materials.

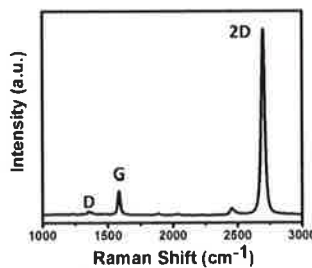
(A) Photon

(B) Electron

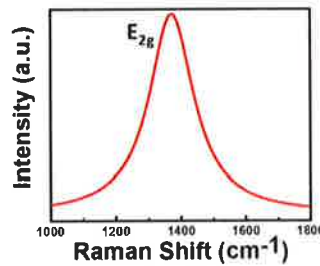
(C) Neutron

(D) Phonon

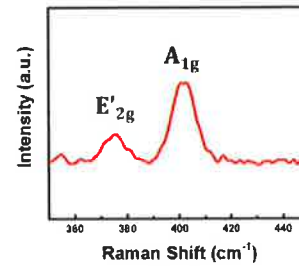
46. Typical Raman spectra of three two-dimensional (2D) materials are presented in the figure below. Identify the 2D materials that corresponds to its Raman fingerprint.



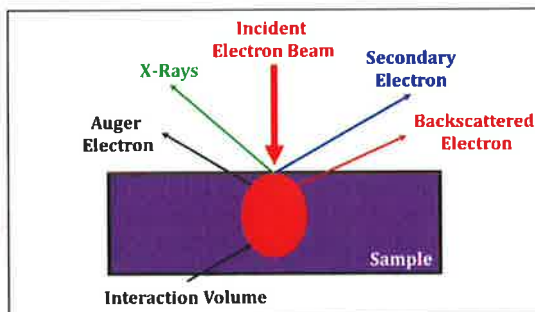
Graphene



h-BN

 MoS_2

47. The electron-solid interactions in a scanning electron microscope (SEM) is described in the following figure. Explain the origin of secondary electrons and X-rays?



Answer: Secondary e^- are the e^- that are kicked out of the outer shells on incident e^- beam.

X-rays are originated when the I.B. hits the core shell e^- of that e^- gets excited itself in the higher level shell.

Name:

UIN:

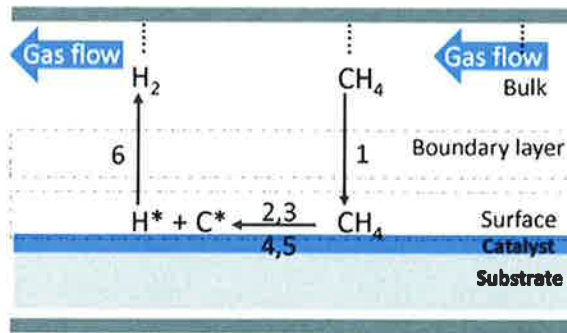
48. Write down the steps required to transfer graphene onto SiO_2/Si substrates from the Cu foil via 'carrier film method'.

Answer: - PMMA spin coating.

- Etching Cu foil.

- Fishing for graphene using SiO_2/Si substrate.

49. Role of kinetic factors for the synthesis of graphene on Cu. State the steps (1-6).



① Diffusion

② Absorption

③ Reaction.

④ Breakdown of $\text{CH}_4 \rightarrow \text{H}^* + \text{C}^*$

⑤ Desorption of H^*

⑥ Diffusion.

Answer:

~~Diffusion.~~

~~① Diffusion~~

~~Absorption.~~

~~② Breakdown of CH_4 to $\text{C}^* + \text{H}^*$~~

~~Reaction~~

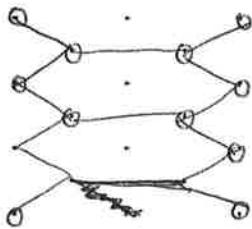
~~③ Absorption~~

~~Desorption.~~

~~④~~

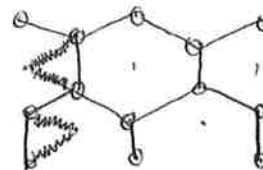
~~Diffusion.~~

50. Crystal Structure = Lattice + Basis. Construct the graphene crystal structure from the following lattice and 2-atom basis.



Lattice

—



Basis

